

Probability basics

Measuring how likely something is.

THE IDEA

Probability is a number between **0** (impossible) and **1** (certain) that measures how likely an event is.

$$P(\text{event}) = \text{favourable outcomes} / \text{total outcomes}$$

THE THREE RULES YOU NEED

Rule	Formula	When
AND	$P(A \text{ and } B) = P(A) \times P(B)$	Independent events — one does not affect the other
OR	$P(A \text{ or } B) = P(A) + P(B)$	Events that cannot both happen
OR (overlapping)	$P(A) + P(B) - P(A \text{ and } B)$	Events that can both happen — subtract the double count
Complement	$P(\text{not } A) = 1 - P(A)$	When "at least one" is easier counted backwards

The complement rule is the most useful shortcut in the whole topic. "At least one head in two coins" would need three cases counted directly; backwards it is one calculation: $1 - P(\text{no heads}) = 1 - \frac{1}{4} = \frac{3}{4}$.

WORKED EXAMPLES

TOPIC 2 — PROBABILITY BASICS

measuring how likely something is

THE DEFINITION

$$P(\text{event}) = \text{favourable outcomes} / \text{total outcomes}$$

always between 0 (impossible) and 1 (certain)

EXAMPLE 1 — one die

total outcomes : 1, 2, 3, 4, 5, 6 → 6

$$P(\text{rolling a 4}) = 1 / 6 \approx 0.167$$

$$P(\text{an even number}) = 3 / 6 = 0.5$$

THE AND RULE — independent events

$$P(A \text{ and } B) = P(A) \times P(B)$$

"independent" = one does not affect the other

example : two coins, both heads

$$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = 0.25$$

THE OR RULE — events that cannot both happen

$$P(A \text{ or } B) = P(A) + P(B)$$

example : a die shows 2 or 5

$$\frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

If they CAN overlap, subtract the overlap :

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

THE COMPLEMENT — the shortcut

$$P(\text{not } A) = 1 - P(A)$$

example : at least one head in two coins

$$P(\text{no heads}) = \frac{1}{4} \rightarrow 1 - \frac{1}{4} = \frac{3}{4}$$

often far easier than counting every winning case

ML LINK : a classifier outputs probabilities, not labels.

They must be between 0 and 1, and sum to 1 across classes.

Why it matters for ML: a classifier does not output a label, it outputs probabilities. They must sit between 0 and 1 and sum to 1 across the classes — which is exactly what the softmax function guarantees.

KEY POINTS FROM THIS TOPIC

Probability	favourable / total, always between 0 and 1
AND (independent)	$P(A) \times P(B)$
OR (exclusive)	$P(A) + P(B)$
OR (overlapping)	$P(A) + P(B) - P(A \text{ and } B)$
Complement	$P(\text{not } A) = 1 - P(A)$

TEST — Topic 2

1. A die is rolled. What is $P(\text{a number greater than 4})$? $\rightarrow 2/6 = 1/3$
2. Two dice are rolled. What is $P(\text{both show 6})$? $\rightarrow 1/6 \times 1/6 = 1/36$
3. Three coins are flipped. What is $P(\text{at least one head})$? $\rightarrow P(\text{no heads}) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = 1/8 \rightarrow 1 - 1/8 = 7/8$